WORKED SOLUTION

STEP I 2007, Question 2

Pure • Inverse Tangent Identities

STEP I 2007, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q2

The key idea

An equation between inverse tangents becomes an equation between ordinary fractions as soon as we take the tangent of both sides and use the addition formula for $\tan(A+B)$. The price is that tan repeats every π, so $\tan(A+B)=1$ only tells us $A+B=\frac{1}{4}\pi+k\pi$; we need a word about the sizes of the angles to pin down k. After that, each part is a small piece of number theory: factorise, or use a divisibility argument.

Solution

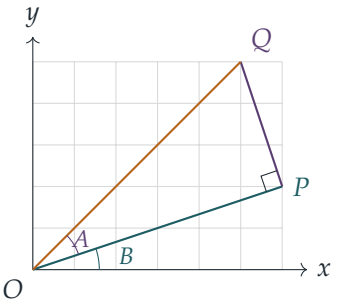
Part (i)

Since $\tan A=\frac{1}{2}$ and $\tan B=\frac{1}{3}$,

$$\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}=\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{6}}=\frac{\frac{5}{6}}{\frac{5}{6}}=1.$$

Now A and B are acute, so $0<A+B<\pi$, and the only angle in $(0,\pi)$ with tangent 1 is $\frac{1}{4}\pi$. Hence $A+B=\frac{1}{4}\pi$, as required.

The picture below shows the same fact on squared paper. With $O=(0,0)$, $P=(6,2)$ and $Q=(5,5)$, the line OP makes angle $\arctan\frac{1}{3}$ with the x-axis; $PQ=(-1,3)$ is perpendicular to $OP=(6,2)$ and half as long, so angle POQ is $\arctan\frac{1}{2}$; and Q lies on the line $y=x$, at $\frac{1}{4}\pi$.



Part (i): the integers p and q

First we rule out $pq = 1$, where the addition formula would divide by zero. Then $p = q = 1$ or $p = q = -1$, giving left-hand sides $\frac{1}{2}\pi$ and $-\frac{1}{2}\pi$, neither of which is $\frac{1}{4}\pi$. So $pq \neq 1$, and taking tangents of both sides of the given equation,

$$\frac{\frac{1}{p} + \frac{1}{q}}{1 - \frac{1}{pq}} = 1.$$

Multiplying top and bottom by pq gives $\dfrac{p+q}{pq-1} = 1$, so

$$p + q = pq - 1 \quad \Longrightarrow \quad pq - p - q + 1 = 2 \quad \Longrightarrow \quad (p-1)(q-1) = 2,$$

which is the required result.

Since $p-1$ and $q-1$ are integers whose product is 2, the pair $(p-1, q-1)$ is one of

$$(1,2), \quad (2,1), \quad (-1,-2), \quad (-2,-1),$$

giving $(p,q) = (2,3)$, $(3,2)$, $(0,-1)$ or $(-1,0)$. The last two are excluded because p and q are non-zero. So $\{p,q\} = \{2,3\}$, and both orders really do satisfy the equation by the first half of part (i).

Ask yourself We only showed that a solution must satisfy $(p-1)(q-1) = 2$. Why do we need to check the candidates afterwards?

Taking tangents loses information: $(p-1)(q-1) = 2$ is equivalent to $\tan(\arctan\frac{1}{p} + \arctan\frac{1}{q}) = 1$, which also allows sums such as $-\frac{3}{4}\pi$. Here the candidate pairs either contain 0 (so $\frac{1}{p}$ is meaningless) or are 2 and 3, which we know work.

Part (ii)

Write $\theta = \arctan\frac{1}{r}$ and $\phi = \arctan\dfrac{s}{s+t}$. Taking tangents of the given equation,

$$\frac{\frac{1}{r} + \frac{s}{s+t}}{1 - \frac{s}{r(s+t)}} = 1.$$

The denominator is non-zero: since $r \geq 1$ and $t \geq 1$, we have $r(s+t) > s$. Multiplying top and bottom by $r(s+t)$,

$$\frac{(s+t)+rs}{r(s+t)-s} = 1 \quad \Longrightarrow \quad s+t+rs = rs+rt-s \quad \Longrightarrow \quad t(r-1) = 2s.$$

Ask yourself How does the condition on the highest common factor of s and t come in?

The equation $t(r-1) = 2s$ says that t divides $2s$. As $\text{hcf}(s,t) = 1$, t shares no prime factor with s, so every prime factor of t must come from the 2: t divides 2. Since t is a positive integer, the only possible values are

$$t = 1 \quad \text{or} \quad t = 2.$$

- If $t = 1$: $r - 1 = 2s$, so $r = 2s + 1$.
- If $t = 2$: $2(r-1) = 2s$, so $r = s + 1$. (Here s must be odd, since $\text{hcf}(s,2) = 1$.)

Both cases genuinely occur. Indeed $0 < \frac{1}{r} \leq 1$ and $0 < \frac{s}{s+t} < 1$, so θ and ϕ both lie in $(0, \frac{1}{4}\pi]$ and $0 < \theta + \phi < \frac{1}{2}\pi$. Reversing the algebra above, $t(r-1) = 2s$ gives $\tan(\theta+\phi) = 1$, and so $\theta + \phi = \frac{1}{4}\pi$ exactly. For example, $t = 1,\ s = 1$ gives $\arctan\frac{1}{3} + \arctan\frac{1}{2}$, and $t = 2,\ s = 1$ gives $\arctan\frac{1}{2} + \arctan\frac{1}{3}$: the identity of part (i) both ways round.

Checks

- On a calculator, $\arctan 0.5 + \arctan(1/3) = 0.4636 + 0.3218 = 0.7854 = \frac{1}{4}\pi$.
- A computer search over all non-zero p, q with $|p|, |q| \leq 200$ finds only $(2,3)$ and $(3,2)$.
- A search over $r < 400$ and $s, t < 150$ with $\text{hcf}(s,t) = 1$ finds only $t = 1,\ r = 2s+1$ and $t = 2,\ r = s+1$; for instance $\arctan\frac{1}{8} + \arctan\frac{7}{9} = \frac{1}{4}\pi$ ($s = 7,\ t = 2$) to machine precision.

Common slips

- Concluding $A + B = \frac{1}{4}\pi$ from $\tan(A+B) = 1$ without saying why $A + B$ cannot be $\frac{5}{4}\pi$.
- Forgetting the negative factorisations $(-1)(-2)$ and $(-2)(-1)$ of 2, or forgetting to discard them because they give a zero.
- Saying "t divides $2s$, so t divides 2" without quoting $\text{hcf}(s,t) = 1$: this is exactly where the hypothesis is needed.

Beyond STEP

The addition formula for tan is really multiplication of complex numbers: $\arctan \frac{1}{p}$ is the argument of $p + \mathrm{i}$, so part (i) is the statement that $(2+\mathrm{i})(3+\mathrm{i}) = 5 + 5\mathrm{i}$ has argument $\frac{1}{4}\pi$. Identities like Machin's $4\arctan\frac{1}{5} - \arctan\frac{1}{239} = \frac{1}{4}\pi$, once used to compute π to hundreds of digits, come from factorising Gaussian integers in exactly this way. Needham's *Visual Complex Analysis* develops the geometry of complex multiplication beautifully, and Davenport's *The Higher Arithmetic* covers the divisibility arguments used in part (ii).

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.